I had already deployed my code onto Railway in Project 2 and included basically all of the functionality required in Project 3 so I am pretty much done with the whole project. The only functionality I wanted to add was the ability to return a specific entry using its ID.

Here are my objects: Customer(customerId, firstName, lastName, email, phone, address); Category (categoryId, name, description); Order (orderId, customerId, orderDate, totalAmount, status); OrderItem (orderItemId, orderId, productId, quantity, unitPrice); Payment (paymentId, orderId, paymentDate, amount, paymentMethod); Product (productId, name, description, price, categoryId); and Shipment (shipmentId, orderId, address, status). I have implemented a Database, a Data layer, a business layer, and a service layer. The service layer has getAll, getOne, create, update, and deleteOne implemented for all objects. I have implemented a client layer which gives the client the ability to view records from any table, insert new records, edit old ones, and delete any. I've include a picture of my current frontend for the Customers Table. I have a single dashboard.html file that holds all the code for every table and allows the client to select between tables. I've included the code for the dashboard.html as a file.

I need to add functionality to each table that allows the client to input an ID and "search" the table for that ID and return the result if there is any. Since no objects share the same ID, this will never return more than 1 object and we can use the getOne method that is defined for every object in their controllers to handle the getting of them. At the top of the screen, just to the right of the "Add Customer" button, I want a text field with an accompanying "Search" button that allows the user to input an ID and hit search. This should either return the single found entry, or an empty table. If the text field is empty, it should return all results back to the screen. Can you write the code for this?

This allowed me to update the dashboard with the added buttons and search logic. I had to give Gemini the code from my MainApp before it was able to update my routes effectively. This was all the code I needed to change for Project 4.

Next, I will discuss deployment.

🚀 Deployment Guide

Follow these steps to set up the environment and run the **Online Commerce App** on your local machine.

1. Prerequisites & Repository Setup

- **Source Code**: Download or clone the GitHub repository. All necessary Java source code is located within the `online-commerce-app` folder.

- **Environment**: The developer recommends using **Visual Studio Code** with the appropriate Java extensions. This allows you to run files simply by clicking the **Play** arrow in the top-right corner.

2. Database Configuration (MySQL Workbench)

Before running the application, you must initialize the data layer:

- **Connection**: Open MySQL Workbench and create a new connection. Use any name you prefer, but ensure it is locally hosted on **port 3306**.
- **Initialization**: Locate the `db_creation.sql` and `db_insert.sql` files in the root directory. Run both scripts to create and populate a database named `railway`.

3. Application Configuration

You must link the Java source code to your local database instance:

- **Database Utility**: Navigate to `src/main/java/com/commerce/util/DatabaseUtil.java`.
- **Connection String**: Update the `url` variable to: `jdbc:mysql://localhost:3306/railway?allowPublicKeyRetrieval=true&useSSL=false`.
- **Credentials**: Ensure the `username` and `password` variables match the environment you created in MySQL Workbench.

4. Testing & Execution

Once the configuration is complete, you can verify the layers of the application:

- **Verify Data/Business Layers**: Navigate to `src/main/java/com/commerce/test` and run the `DatabaseHarness` and `BusinessHarness` java files.
- **Launch the Application**: Open `src/main/java/com/commerce/app/MainApp.java` and run the file. This serves as the engine for the entire project and will host the app locally.

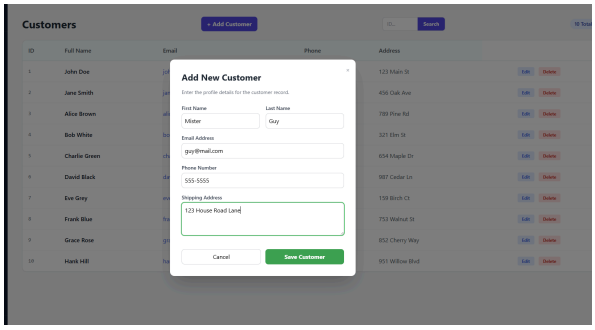
5. Accessing the Dashboard

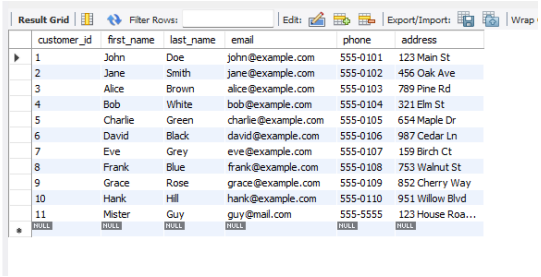
While **MainApp.java** is running, you can interact with the client layer:

- **URL**: Navigate to http://localhost:7070/dashboard in your web browser.
- **Functionality**: From this interface, you can view all tables, insert new records, and edit or delete existing entries.

Finally, here are the screenshots for functionality.

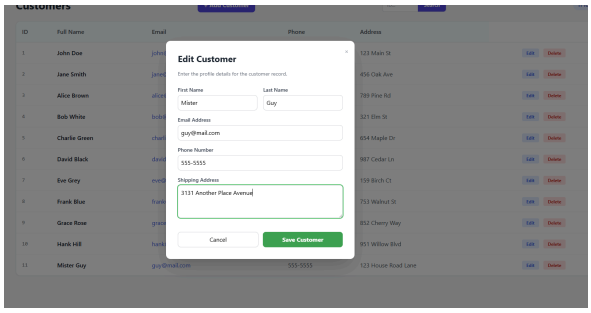
Customer (Insert)

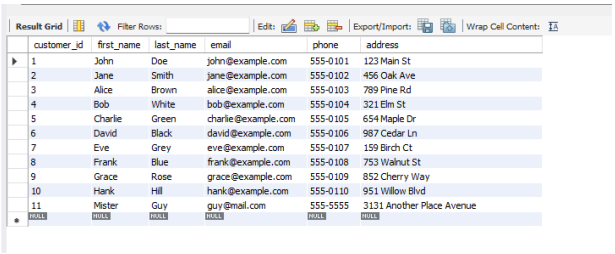




customer_id	first_name	last_name	email	phone	address
1	John	Doe	john@example.com	555-0101	123 Main St
2	Jane	Smith	jane@example.com	555-0102	456 Oak Ave
3	Alice	Brown	alice@example.com	555-0103	789 Pine Rd
4	Bob	White	bob@example.com	555-0104	321 Elm St
5	Charlie	Green	charlie@example.com	555-0105	654 Maple Dr
6	David	Black	david@example.com	555-0106	987 Cedar Ln
7	Eve	Grey	eve@example.com	555-0107	159 Birch Ct
8	Frank	Blue	frank@example.com	555-0108	753 Walnut St
9	Grace	Rose	grace@example.com	555-0109	852 Cherry Way
10	Hank	Hill	hank@example.com	555-0110	951 Willow Blvd
11	Mister	Guy	guy@mail.com	555-5555	123 House Roa...
NULL	NULL	NULL	NULL	NULL	NULL

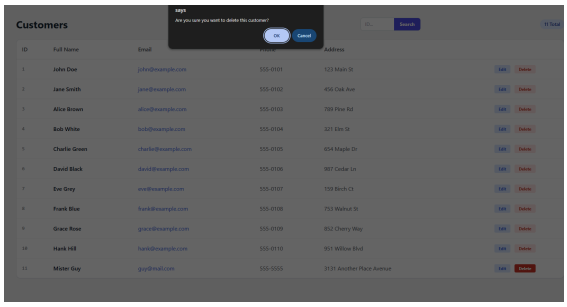
Customer (Update)

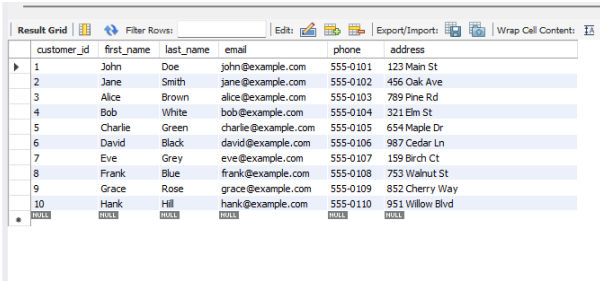




customer_id	first_name	last_name	email	phone	address
1	John	Doe	john@example.com	555-0101	123 Main St
2	Jane	Smith	jane@example.com	555-0102	456 Oak Ave
3	Alice	Brown	alice@example.com	555-0103	789 Pine Rd
4	Bob	White	bob@example.com	555-0104	321 Elm St
5	Charlie	Green	charlie@example.com	555-0105	654 Maple Dr
6	David	Black	david@example.com	555-0106	987 Cedar Ln
7	Eve	Grey	eve@example.com	555-0107	159 Birch Ct
8	Frank	Blue	frank@example.com	555-0108	753 Walnut St
9	Grace	Rose	grace@example.com	555-0109	852 Cherry Way
10	Hank	Hill	hank@example.com	555-0110	951 Willow Blvd
11	Mister	Guy	guy@mail.com	555-5555	3131 Another Place Avenue
NULL	NULL	NULL	NULL	NULL	NULL

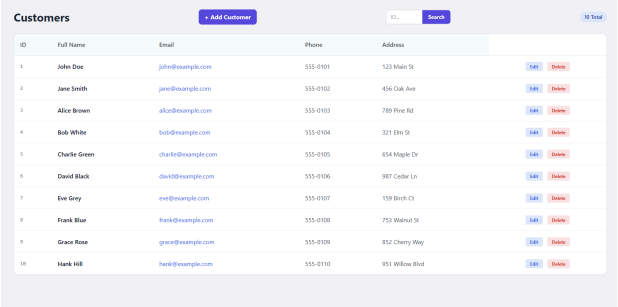
Customer (Delete)

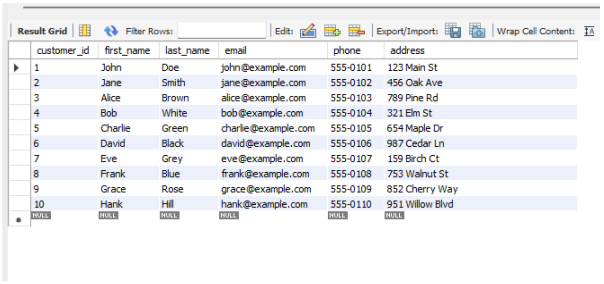




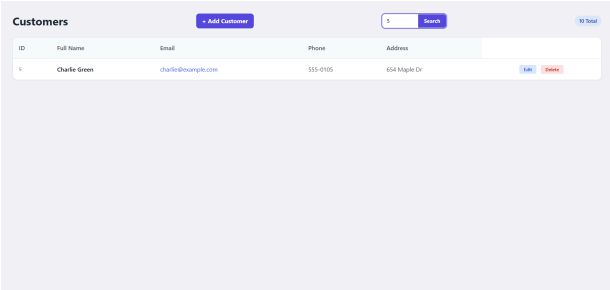
customer_id	first_name	last_name	email	phone	address
1	John	Doe	john@example.com	555-0101	123 Main St
2	Jane	Smith	jane@example.com	555-0102	456 Oak Ave
3	Alice	Brown	alice@example.com	555-0103	789 Pine Rd
4	Bob	White	bob@example.com	555-0104	321 Elm St
5	Charlie	Green	charlie@example.com	555-0105	654 Maple Dr
6	David	Black	david@example.com	555-0106	987 Cedar Ln
7	Eve	Grey	eve@example.com	555-0107	159 Birch Ct
8	Frank	Blue	frank@example.com	555-0108	753 Walnut St
9	Grace	Rose	grace@example.com	555-0109	852 Cherry Way
10	Hank	Hill	hank@example.com	555-0110	951 Willow Blvd
NULL	NULL	NULL	NULL	NULL	NULL

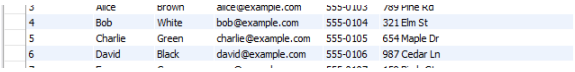
Customer (Get All)



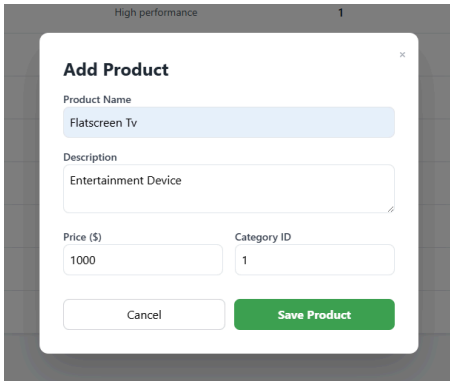


Customer (Get One)

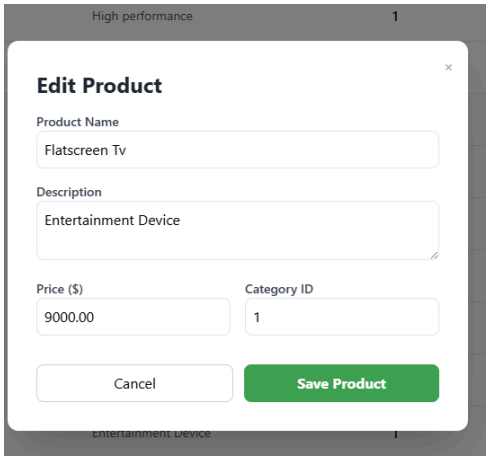


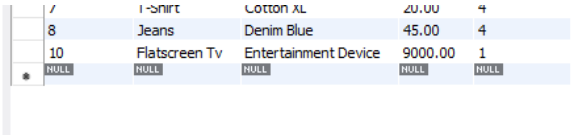


Product (Insert)



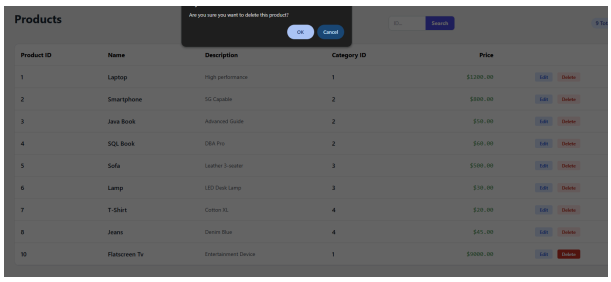
Products (Update)

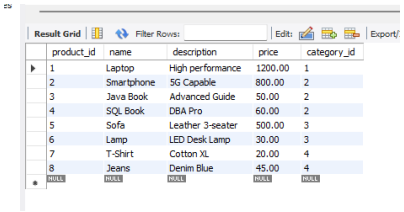




7	T-Shirt	Cotton XL	20.00	4
8	Jeans	Denim Blue	45.00	4
10	Flatscreen Tv	Entertainment Device	9000.00	1
NULL	NULL	NULL	NULL	NULL

Products (Delete)

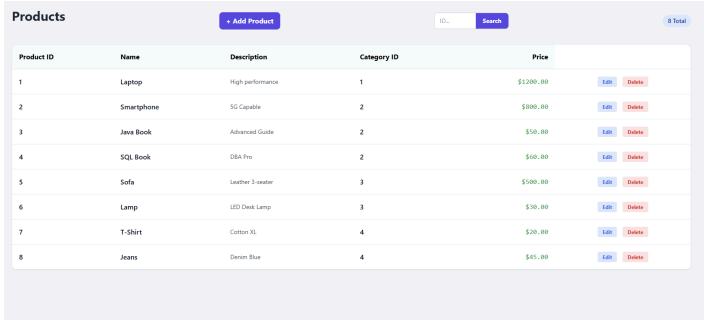




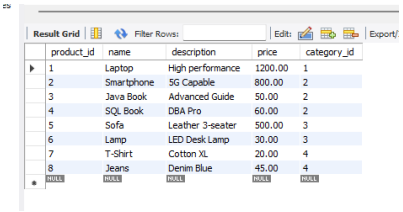
Result Grid | Filter Rows: | Edit: | Export/

product_id	name	description	price	category_id
1	Laptop	High performance	1200.00	1
2	Smartphone	5G Capable	800.00	2
3	Java Book	Advanced Guide	50.00	2
4	SQL Book	DBA Pro	60.00	2
5	Sofa	Leather 3-seater	500.00	3
6	Lamp	LED Desk Lamp	30.00	3
7	T-Shirt	Cotton XL	20.00	4
8	Jeans	Denim Blue	45.00	4
NULL	NULL	NULL	NULL	NULL

Products (Get All)



Product ID	Name	Description	Category ID	Price	
1	Laptop	High performance	1	$1200.00	Edit Delete
2	Smartphone	5G Capable	2	$800.00	Edit Delete
3	Java Book	Advanced Guide	2	$50.00	Edit Delete
4	SQL Book	DBA Pro	2	$60.00	Edit Delete
5	Sofa	Leather 3-seater	3	$500.00	Edit Delete
6	Lamp	LED Desk Lamp	3	$30.00	Edit Delete
7	T-Shirt	Cotton XL	4	$20.00	Edit Delete
8	Jeans	Denim Blue	4	$45.00	Edit Delete



Result Grid | Filter Rows: | Edit: | Export/

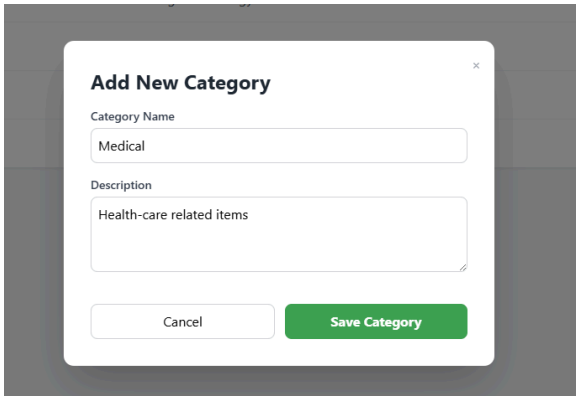
product_id	name	description	price	category_id
1	Laptop	High performance	1200.00	1
2	Smartphone	5G Capable	800.00	2
3	Java Book	Advanced Guide	50.00	2
4	SQL Book	DBA Pro	60.00	2
5	Sofa	Leather 3-seater	500.00	3
6	Lamp	LED Desk Lamp	30.00	3
7	T-Shirt	Cotton XL	20.00	4
8	Jeans	Denim Blue	45.00	4
NULL	NULL	NULL	NULL	NULL

Products (Get One)

Product ID	Name	Description	Category ID	Price	
6	Lamp	LED Desk Lamp	3	$30.00	Edit Delete

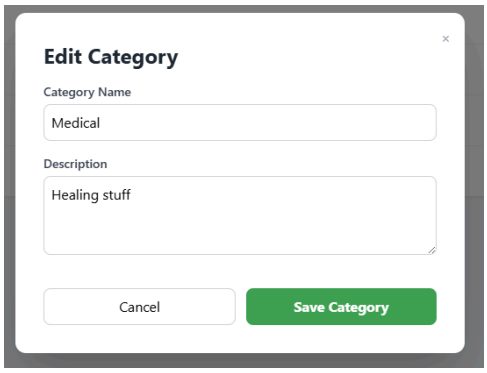
5	Sofa	Leather 3-seater	500.00	3
6	Lamp	LED Desk Lamp	30.00	3
7	T-Shirt	Cotton XL	20.00	4

Categories (Insert)



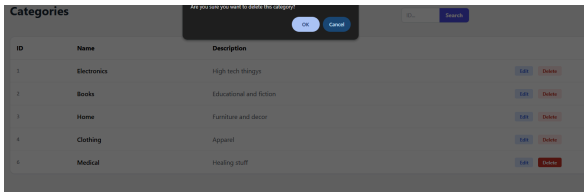
category_id	name	description
1	Electronics	High tech thingys
2	Books	Educational and fiction
3	Home	Furniture and decor
4	Clothing	Apparel
6	Medical	Health-care related items
NULL	NULL	NULL

Categories (Update)



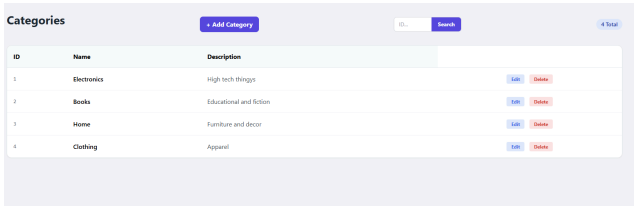
4	Clothing	Apparel
6	Medical	Healing stuff
NULL	NULL	NULL

Categories (Delete)



category_id	name	description
1	Electronics	High tech thingys
2	Books	Educational and fiction
3	Home	Furniture and decor
4	Clothing	Apparel
NULL	NULL	NULL

Categories (Get All)

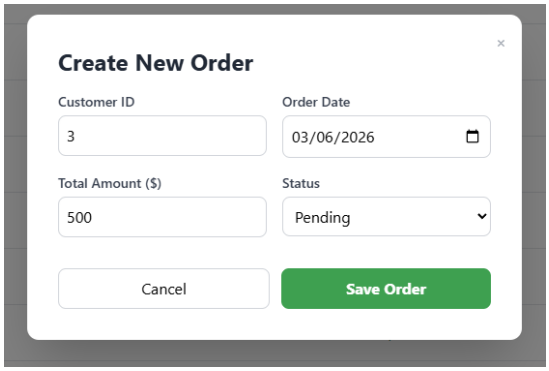


category_id	name	description
1	Electronics	High tech thingys
2	Books	Educational and fiction
3	Home	Furniture and decor
4	Clothing	Apparel
NULL	NULL	NULL

Categories (Get One)

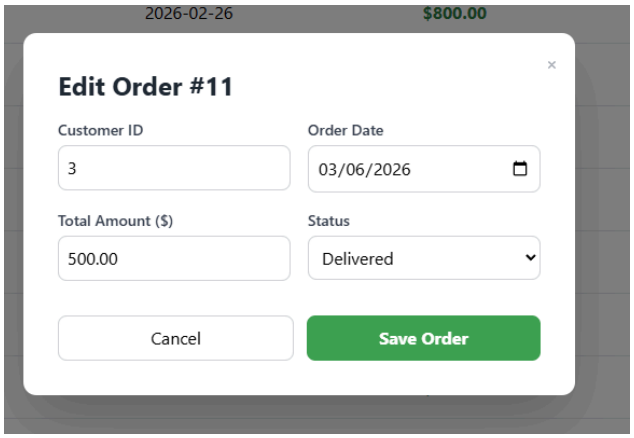
2	Books	Educational and fiction
3	Home	Furniture and decor
4	Clothing	Apparel

Orders (Insert)



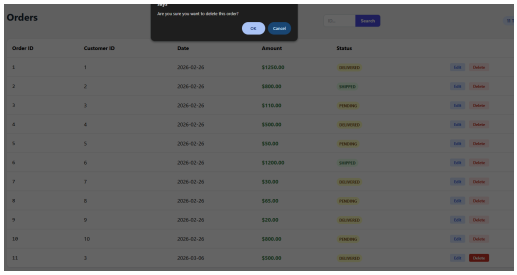
order_id	customer_id	order_date	total_amount	status
1	1	2026-02-26	1250.00	Delivered
2	2	2026-02-26	800.00	Shipped
3	3	2026-02-26	110.00	Pending
4	4	2026-02-26	500.00	Delivered
5	5	2026-02-26	50.00	Pending
6	6	2026-02-26	1200.00	Shipped
7	7	2026-02-26	30.00	Delivered
8	8	2026-02-26	65.00	Pending
9	9	2026-02-26	20.00	Delivered
10	10	2026-02-26	800.00	Pending
11	3	2026-03-06	500.00	Pending
NULL	NULL	NULL	NULL	NULL

Orders (Update)



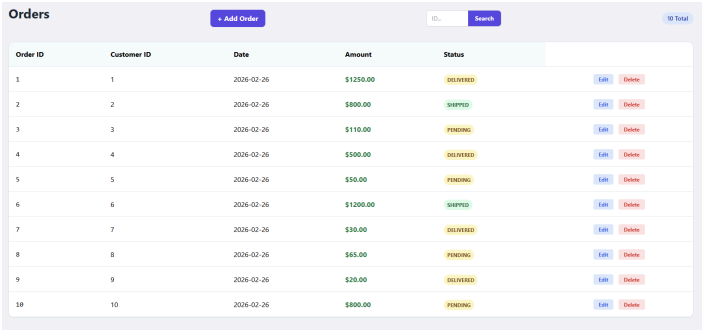
10	10	2026-02-26	800.00	Pending
11	3	2026-03-06	500.00	Delivered
NULL	NULL	NULL	NULL	NULL

Orders (Delete)



order_id	customer_id	order_date	total_amount	status
1	1	2026-02-26	1250.00	Delivered
2	2	2026-02-26	800.00	Shipped
3	3	2026-02-26	110.00	Pending
4	4	2026-02-26	500.00	Delivered
5	5	2026-02-26	50.00	Pending
6	6	2026-02-26	1200.00	Shipped
7	7	2026-02-26	30.00	Delivered
8	8	2026-02-26	65.00	Pending
9	9	2026-02-26	20.00	Delivered
10	10	2026-02-26	800.00	Pending
NULL	NULL	NULL	NULL	NULL

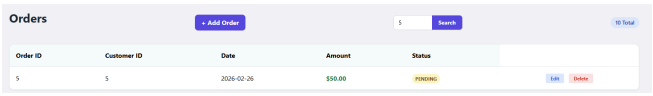
Orders (Get All)



Result Grid | Filter Rows: | Edit:

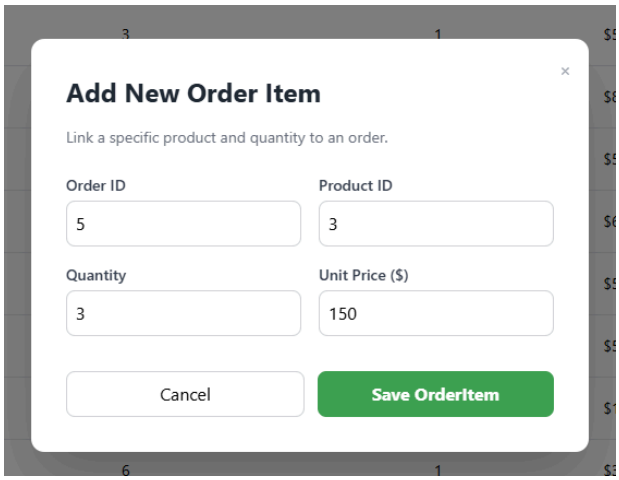
order_id	customer_id	order_date	total_amount	status
1	1	2026-02-26	1250.00	Delivered
2	2	2026-02-26	800.00	Shipped
3	3	2026-02-26	110.00	Pending
4	4	2026-02-26	500.00	Delivered
5	5	2026-02-26	50.00	Pending
6	6	2026-02-26	1200.00	Shipped
7	7	2026-02-26	30.00	Delivered
8	8	2026-02-26	65.00	Pending
9	9	2026-02-26	20.00	Delivered
10	10	2026-02-26	800.00	Pending
NULL	NULL	NULL	NULL	NULL

Orders (Get One)



5	5	2026-02-26	50.00	Pending
6	6	2026-02-26	1200.00	Shipped

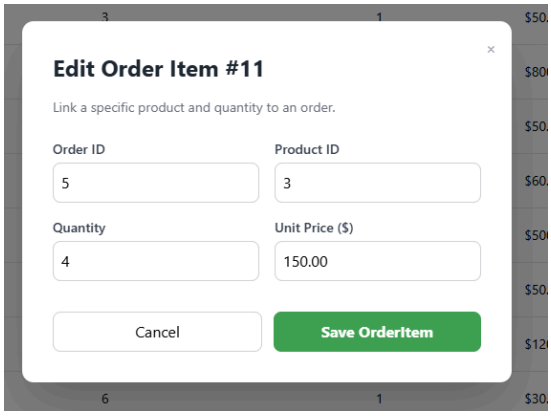
Order Items (Insert)



Result Grid | Filter Rows: | Edit:

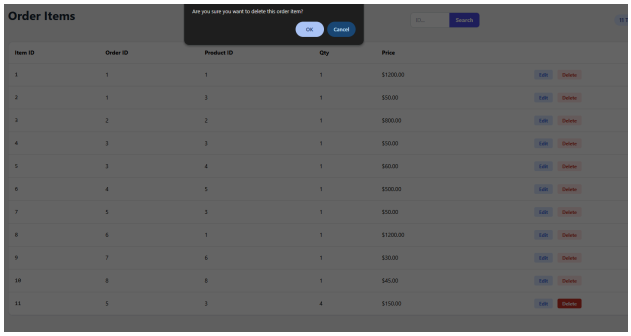
order_item_id	order_id	product_id	quantity	unit_price
1	1	1	1	1200.00
2	1	3	1	50.00
3	2	2	1	800.00
4	3	3	1	50.00
5	3	4	1	60.00
6	4	5	1	500.00
7	5	3	1	50.00
8	6	1	1	1200.00
9	7	6	1	30.00
10	8	8	1	45.00
11	5	3	3	150.00
NULL	NULL	NULL	NULL	NULL

Order Items (Update)



10	8	8	1	45.00
11	5	3	4	150.00
NULL	NULL	NULL	NULL	NULL

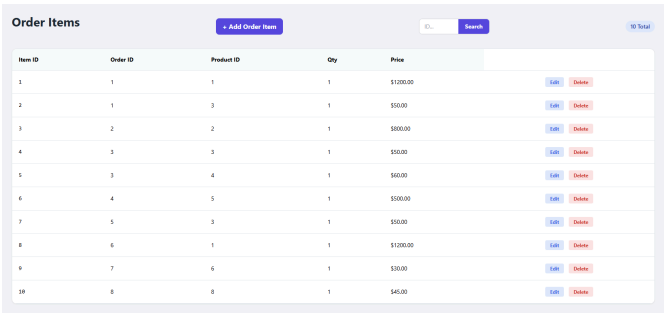
Order Items (Delete)



Result Grid | Filter Rows: | Edit:

order_item_id	order_id	product_id	quantity	unit_price
1	1	1	1	1200.00
2	1	3	1	50.00
3	2	2	1	800.00
4	3	3	1	50.00
5	3	4	1	60.00
6	4	5	1	500.00
7	5	3	1	50.00
8	6	1	1	1200.00
9	7	6	1	30.00
10	8	8	1	45.00
NULL	NULL	NULL	NULL	NULL

Order Items (Get All)



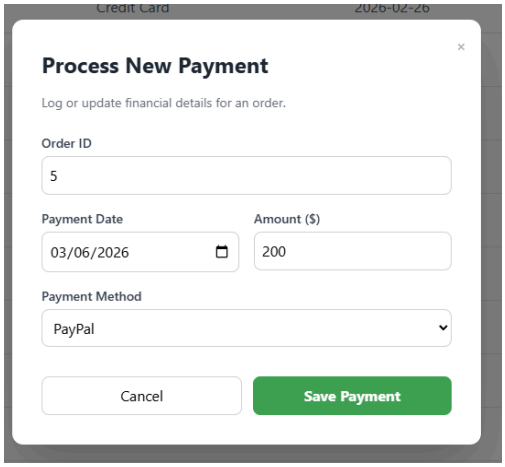
Result Grid | Filter Rows: | Edit:

order_item_id	order_id	product_id	quantity	unit_price
1	1	1	1	1200.00
2	1	3	1	50.00
3	2	2	1	800.00
4	3	3	1	50.00
5	3	4	1	60.00
6	4	5	1	500.00
7	5	3	1	50.00
8	6	1	1	1200.00
9	7	6	1	30.00
10	8	8	1	45.00
NULL	NULL	NULL	NULL	NULL

Order Items (Get One)

3	2	2	1	800.00
4	3	3	1	50.00
5	3	4	1	60.00

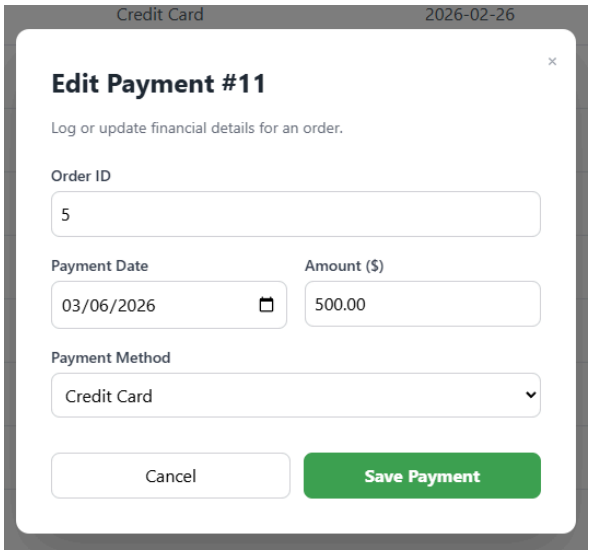
Payments (Insert)



Result Grid | Filter Rows: | Edit:

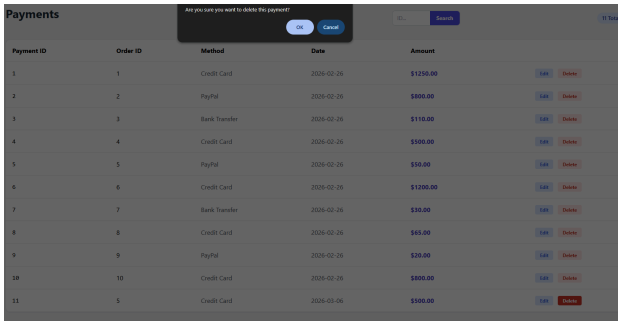
payment_id	order_id	payment_date	amount	payment_method
1	1	2026-02-26	1250.00	Credit Card
2	2	2026-02-26	800.00	PayPal
3	3	2026-02-26	110.00	Bank Transfer
4	4	2026-02-26	500.00	Credit Card
5	5	2026-02-26	50.00	PayPal
6	6	2026-02-26	1200.00	Credit Card
7	7	2026-02-26	30.00	Bank Transfer
8	8	2026-02-26	65.00	Credit Card
9	9	2026-02-26	20.00	PayPal
10	10	2026-02-26	800.00	Credit Card
11	5	2026-03-06	200.00	PayPal
NULL	NULL	NULL	NULL	NULL

Payments (Update)



10	10	2026-02-26	800.00	Credit Card
11	5	2026-03-06	500.00	Credit Card
NULL	NULL	NULL	NULL	NULL

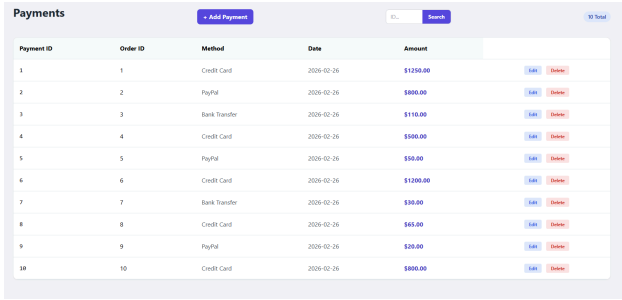
Payments (Delete)



Result Grid | Filter Rows: | Edit:

payment_id	order_id	payment_date	amount	payment_method
1	1	2026-02-26	1250.00	Credit Card
2	2	2026-02-26	800.00	PayPal
3	3	2026-02-26	110.00	Bank Transfer
4	4	2026-02-26	500.00	Credit Card
5	5	2026-02-26	50.00	PayPal
6	6	2026-02-26	1200.00	Credit Card
7	7	2026-02-26	30.00	Bank Transfer
8	8	2026-02-26	65.00	Credit Card
9	9	2026-02-26	20.00	PayPal
10	10	2026-02-26	800.00	Credit Card
NULL	NULL	NULL	NULL	NULL

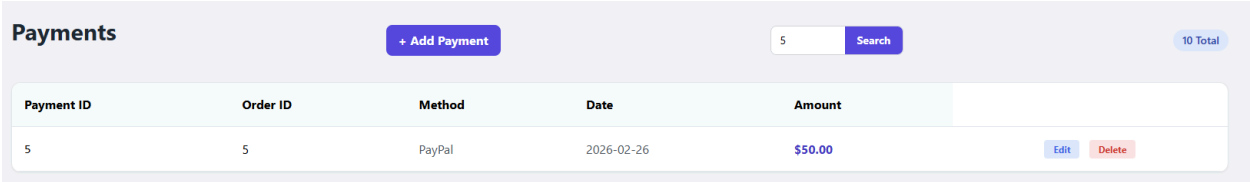
Payments (Get All)



Result Grid | Filter Rows: | Edit:

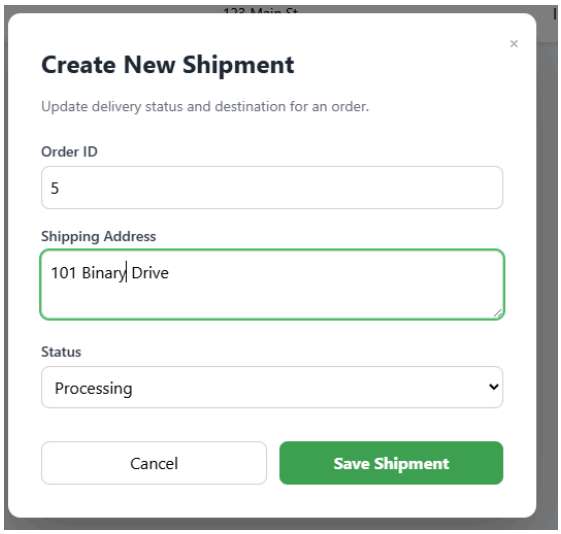
payment_id	order_id	payment_date	amount	payment_method
1	1	2026-02-26	1250.00	Credit Card
2	2	2026-02-26	800.00	PayPal
3	3	2026-02-26	110.00	Bank Transfer
4	4	2026-02-26	500.00	Credit Card
5	5	2026-02-26	50.00	PayPal
6	6	2026-02-26	1200.00	Credit Card
7	7	2026-02-26	30.00	Bank Transfer
8	8	2026-02-26	65.00	Credit Card
9	9	2026-02-26	20.00	PayPal
10	10	2026-02-26	800.00	Credit Card
NULL	NULL	NULL	NULL	NULL

Payments (Get One)



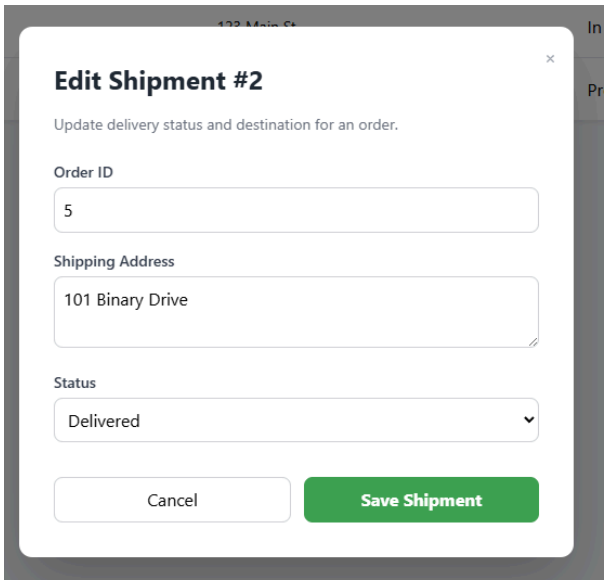
4	4	2026-02-26	500.00	Credit Card
5	5	2026-02-26	50.00	PayPal
6	6	2026-02-26	1200.00	Credit Card

Shipments (Insert)



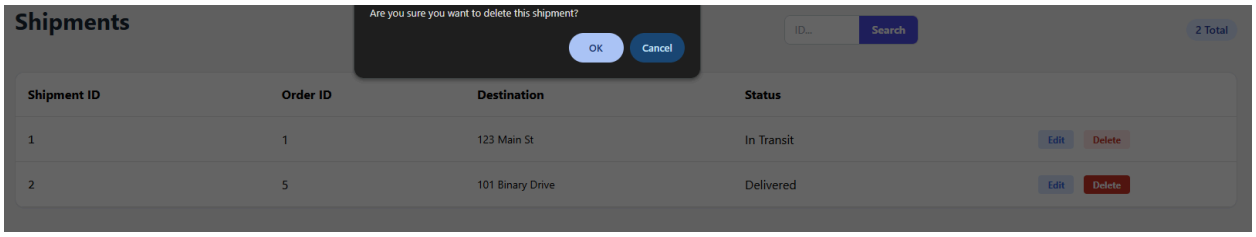
shipment_id	order_id	address	status
1	1	123 Main St	In Transit
2	5	101 Binary Drive	Processing
NULL	NULL	NULL	NULL

Shipments (Update)



shipment_id	order_id	address	status
1	1	123 Main St	In Transit
2	5	101 Binary Drive	Delivered
NULL	NULL	NULL	NULL

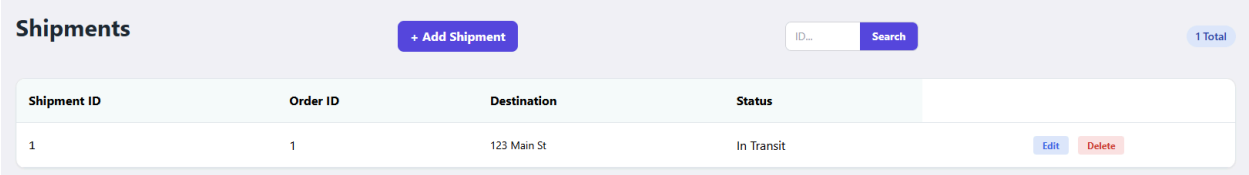
Shipments (Delete)



Result Grid | Filter Rows: | Edit:

shipment_id	order_id	address	status
1	1	123 Main St	In Transit
NULL	NULL	NULL	NULL

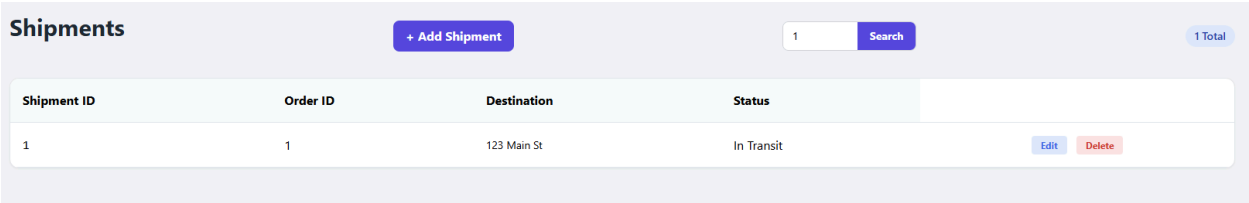
Shipments (Get All)

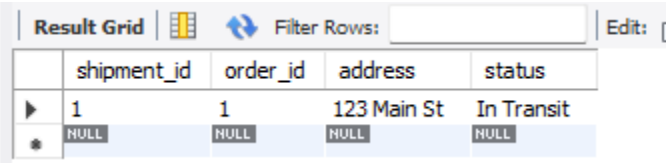


Result Grid | Filter Rows: | Edit:

shipment_id	order_id	address	status
1	1	123 Main St	In Transit
NULL	NULL	NULL	NULL

Shipments (Get One)





shipment_id	order_id	address	status
1	1	123 Main St	In Transit
NULL	NULL	NULL	NULL